Letter Sequences & Alphabet Code

Year 4 · **Homework Worksheet**

Worked Example

Worked example: If the code for PEN is QFO (each letter shifts forward by 1 in the alphabet), what is the code for BUS?

Answer: Shift every letter of BUS forward by 1: B→C, U→V, S→T. Answer: CVT.

Easy / Standard (15 questions)

1. What comes next in the sequence: D, E, F, G, ?
2. What comes next in the sequence: Q, R, S, T, ?
3. What comes next in the sequence: R, S, T, U, ?
4. What comes next in the sequence: S, T, U, V, ?
5. What comes next in the sequence: T, U, V, W, ?
6. What comes next in the sequence: U, V, W, X, ?
7. What comes next in the sequence: V, W, X, Y, ?
8. What comes next in the sequence: E, D, C, B, ?
9. What comes next in the sequence: F, E, D, C, ?
10. What comes next in the sequence: G, F, E, D, ?
11. What comes next in the sequence: H, G, F, E, ?
12. What comes next in the sequence: I, H, G, F, ?
13. What comes next in the sequence: J, I, H, G, ?
14. What comes next in the sequence: K, J, I, H, ?
15. What comes next in the sequence: L, K, J, I, ?

Standard (15 questions)

1. What comes next in the sequence: P, R, T, V, ?
2. What comes next in the sequence: Q, S, U, W, ?
3. What comes next in the sequence: R, T, V, X, ?
4. What comes next in the sequence: A, D, G, J, ?
5. What comes next in the sequence: B, E, H, K, ?
6. What comes next in the sequence: C, F, I, L, ?
7. What comes next in the sequence: D, G, J, M, ?
8. What comes next in the sequence: E, H, K, N, ?
9. What comes next in the sequence: F, I, L, O, ?
10. What comes next in the sequence: G, J, M, P, ?

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11. What comes next in the sequence: H, K, N, Q, ?

12. What comes next in the sequence: I, L, O, R, ?

13. What comes next in the sequence: J, M, P, S, ?

14. What comes next in the sequence: K, N, Q, T, ?

15. What comes next in the sequence: L, O, R, U, ?

Standard / Advanced (15 questions)

1. What letter comes immediately after the next term in the sequence: C, F, I, L, ?

2. If the code for CAT is DBU, what letter would N become using the same code?

3. What letter comes immediately after the next term in the sequence: X, U, R, O, ?

4. If the code for DOG is EPH, what letter would S become using the same code? A) R B) S C) T D) U E) V

5. What letter comes immediately after the next term in the sequence: B, F, J, N, ?

6. A pattern of letters moves the same number of places forward through the alphabet each time. The first 4 letters are D, G, J, M. What letter comes two terms further along the pattern after M?

7. A pattern of letters moves the same number of places back through the alphabet each time. The first 4 letters are W, T, Q, N. What letter comes two terms further along the pattern after N?

8. A tutor makes up a code: every letter of a word is moved 2 places forward in the alphabet. Using this code, what coded word would be made from the word BIG? A) DKI B) DKH C) CKI D) DJI E) DLI

9. A tutor makes up a code: every letter of a word is moved 2 places forward in the alphabet. Using this code, what coded word would be made from the word SUN?

10. A message has been coded by moving every letter 1 place forward in the alphabet. The coded word is "GBU". What was the original word before it was coded?

11. Select ALL that apply: a code shifts every letter forward by 3 places, and under this code CAT becomes FDW. A) The shift used is +3. B) Using the same shift, DOG would become GRJ. C) Using the same shift, DOG would become FQI. D) The shift used is -3. E) Using the same shift, BAT would become EDW.

12. In a shift code, the letter F becomes I. Using the same shift, what would the letter P become?

13. A puzzle gives you this pair as a worked example of a code: PEN → RGP. Work out the shift used, then encode the word "BAT" using the same shift. A) DCV B) DCW C) CBV D) DBV E) ECW

14. A puzzle gives you this pair as a worked example of a code: DOG → FQI. Work out the shift used, then decode the word "KPV" using the same shift. A) HMS B) ITS C) INT D) HNS E) IOT

15. A two-step code first shifts every letter forward by 1 place, then swaps the first and last letters of the result. Apply this to the word "CAT" and give the final 3-letter result.

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